

Total marks – 83

Attempt Questions 1 – 6

Questions are NOT of equal value

Answer each question on a SEPARATE piece of paper clearly marked Question 1, Question 2, etc. Each piece of paper must show your name.

Question 1 (14 marks) Use a separate piece of paper**Marks**

a) Solve the following inequalities;

(i) $x^2 + 13x \geq 90$

2

(ii) $\frac{2}{x-3} \geq 4$

3

(ii) $\frac{1}{x} < \frac{1}{x+1}$

3

b) Solve the equation $|x-3| = 2x+1$

3

c) If $x = \frac{7-2\sqrt{2}}{7+2\sqrt{2}}$, find the value of $x + \frac{1}{x}$

3

Question 2 (13 marks) Use a separate piece of papera) (i) Sketch the graphs $y = 3x$ and $y = |2x-5|$ on the same number plane noting any intercepts with the axes and points of intersection.

2

(ii) Hence solve $3x - |2x-5| \geq 0$

1

b) In your own words describe what is meant by “function” when referring to number plane graphs.

2

c) (i) Sketch the function $y = |x+1| - |x-1|$

2

(ii) Determine whether the function is odd, even or neither, giving reasons for your answer.

2

d) Given that $f(x) = \begin{cases} x^2 & , x > 2 \\ \frac{1}{x+1} & , -1 < x \leq 2 \\ 5 & , x \leq -1 \end{cases}$, find;

(i) $f(3) + f(-1)$

2

(ii) the domain of $f(x)$

1

(iii) the range of $f(x)$

1

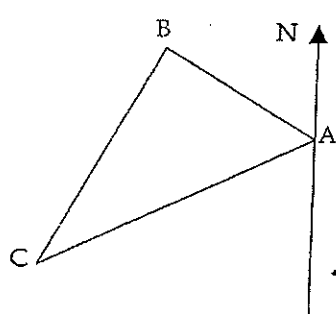
Question 3 (14 marks) Use a *separate* piece of paper

Marks

- a) If $\tan \theta = \frac{5}{12}$ and $\sin \theta < 0$, find the exact value of $\sec \theta$ 2
- b) Solve for θ , correct to the nearest degree where necessary, where $0^\circ \leq \theta \leq 360^\circ$
- (i) $\cos^2 \theta = \cos \theta$ 3
- (ii) $\sqrt{3} \sin \theta - 3 \cos \theta = 0$ 3
- (iii) $\sin 3\theta = \frac{1}{2}$ 3
- (iv) $4 \tan \theta + 1 - \tan^2 \theta = \sec^2 \theta$ 3

Question 4 (13 marks) Use a *separate* piece of paper

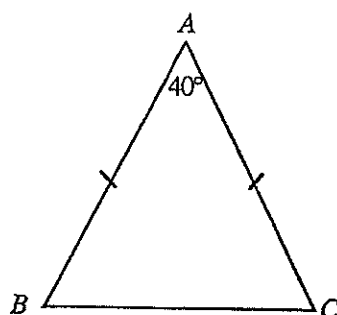
- a) Factorise $(a^2 - b^2)^2 - (a - b)^4$ completely 3



- b) Boat B is 30 nautical miles from harbour A . Another boat C is 37 nautical miles from A and is sailing on a bearing of 245° . The distance between the boats is 34 nautical miles.

- (i) Copy the diagram and indicate the given information 1
- (ii) Find the size of $\angle BCA$ 2
- (iii) Determine the bearing of boat B from boat C , correct to the nearest degree 2

- c) The vertical angle of an isosceles triangle is 40° and its area is 40 cm^2 .



- (i) Show that $AB = \sqrt{80 \operatorname{cosec} 40^\circ}$ 2
- (ii) Hence, or otherwise, calculate the length of BC , correct to one decimal place 3

Question 5 (14 marks) Use a *separate* piece of paper

Marks

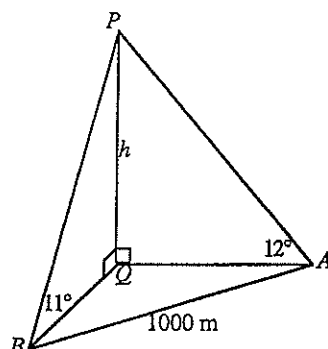
a) (i) On separate diagrams sketch the regions $y > \frac{1}{x}$ and $xy > 1$

3

(ii) Are both regions in (i) the same? In your own words, explain why this happens.

1

b) The angle of elevation of a tower PQ of height h metres, at a point A due East of it, is 12° . From another point B , the bearing of the tower is 051°T and the angle of elevation is 11° . The points A and B are 1000 metres apart and on the same level as the base Q of the tower.



(i) Show that $\angle AQB = 141^\circ$

2

(ii) Consider $\triangle APQ$, show that $AQ = h \tan 78^\circ$

2

(iii) Find a similar expression for BQ

1

(iv) Hence calculate h , correct to the nearest metre.

3

c) Prove that $\frac{1 + \cot \theta}{\operatorname{cosec} \theta} - \frac{1 + \tan \theta}{\sec \theta}$ is independent of θ

2

GO TO THE NEXT PAGE FOR QUESTION 6

Question 6 (15 marks) Use a *separate* piece of paper

Marks

a) Find all the values of x and y , if $\frac{1}{x} + \frac{3}{y} = 3$ and $\frac{1}{x} + y = 1$

3

b) If θ is acute and $\sin \theta = \frac{1}{\sqrt{3}}$

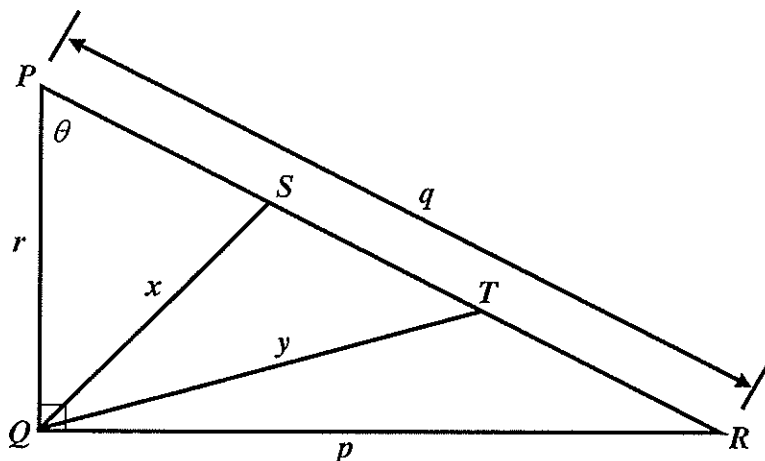
(i) Show that $\frac{\tan \theta}{1 - \sec \theta} = -\sqrt{2} - \sqrt{3}$

3

(ii) Find the value of this fraction when θ is obtuse.

2

c)



The right angled triangle PQR has its hypotenuse PR trisected at the points S and T , that is $PS = ST = TR$.

(i) Show that $\cos \theta = \frac{9r^2 + q^2 - 9x^2}{6qr}$

2

(ii) Show that $\sin \theta = \frac{9p^2 + q^2 - 9y^2}{6pq}$

2

(iii) Hence, or otherwise, deduce that $5q^2 = 9(x^2 + y^2)$

3

Question 1 (14)

a) (i) $x^2 + 13x > 90$

$$x^2 + 13x - 90 > 0$$

$$(x+18)(x-5) > 0$$

$$x < -18 \text{ or } x > 5 \quad (2)$$

(ii) $\frac{2}{x-3} > 4$

$$x-3 \neq 0$$

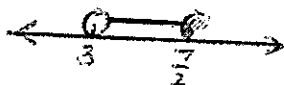
$$x \neq 3$$

$$\frac{2}{x-3} = 4$$

$$2 = 4x - 12$$

$$4x = 14$$

$$x = \frac{7}{2}$$



$$\therefore 3 < x \leq \frac{7}{2} \quad (3)$$

(iii) $\frac{1}{x} < \frac{1}{x+1}$

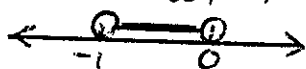
$$x \neq 0, x+1 \neq 0$$

$$x \neq -1$$

$$\frac{1}{x} = \frac{1}{x+1}$$

$$x+1 = x$$

no solutions



$$\therefore -1 < x < 0 \quad (3)$$

b) $|x-3| = 2x+1$

$$x-3 = 2x+1 \text{ OR } -(x-3) = 2x+1$$

$$x = -4$$

$$-x+3 = 2x+1$$

not a solution

$$3x = 2$$

$$x = \frac{2}{3}$$

$$\therefore x = \frac{2}{3} \quad (3)$$

c) $x + \frac{1}{x}$

$$= \frac{7-2\sqrt{2}}{7+2\sqrt{2}} + \frac{7+2\sqrt{2}}{7-2\sqrt{2}}$$

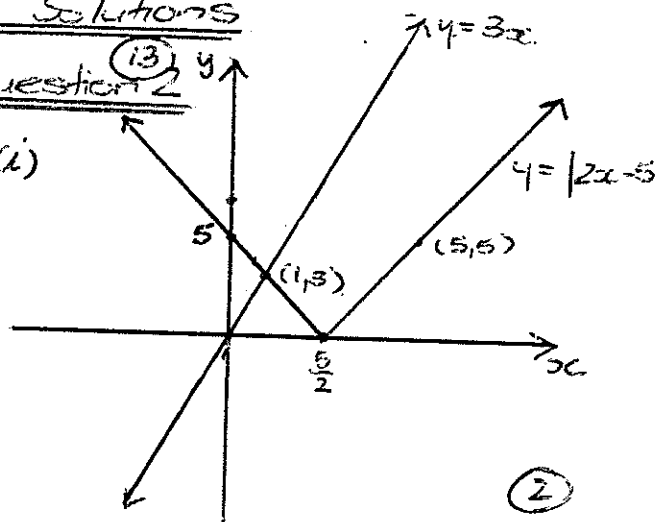
$$= \frac{(7-2\sqrt{2})^2 + (7+2\sqrt{2})^2}{(7+2\sqrt{2})(7-2\sqrt{2})}$$

$$= \frac{2(7^2 + (2\sqrt{2})^2)}{7^2 - (2\sqrt{2})^2}$$

$$= \frac{114}{41} \quad (3)$$

Question 2 (13)

a) (i)



(ii) $3x - |2x-5| \geq 0$

$$3x \geq |2x-5|$$

$$x \geq 1 \quad (1)$$

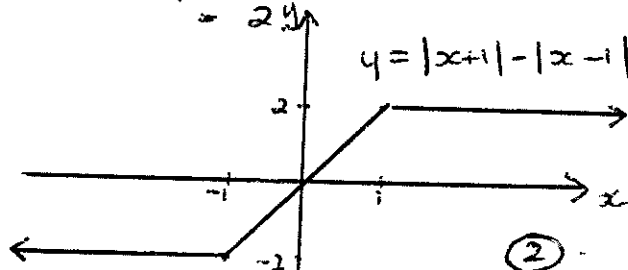
b) A function is a relation that has a maximum of one y value for every x value in its domain. (2)

c) $y = |x+1| - |x-1|$

$$x < -1, y = -1-x - (1-x) = -2$$

$$-1 \leq x \leq 1, y = x+1 - (1-x) = 2x$$

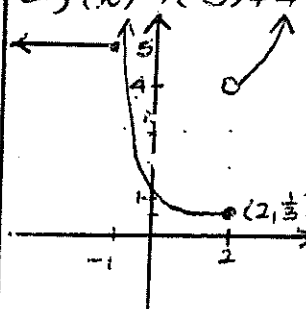
$$x > 1, y = x+1 - (x-1) = 2$$



(ii) $f(-x) = |-x+1| - |-x-1|$
 $= |-1||x-1| - |-1||x+1|$
 $= |x-1| - |x+1|$
 $= -f(x)$

$\therefore$ function is odd (2)

d) (i) $f(3) + f(-1) = 3^2 + 5 = 14 \quad (2)$



(ii) domain: all real x (1)

(iii) range: $y \geq \frac{1}{3} \quad (1)$

Question 3 (14)

a) $\tan \theta = \frac{5}{12}$, $\sin \theta < 0 \Rightarrow Q3$



$\sec \theta = -\frac{13}{12}$ (2)

b) (i) $\cos^2 \theta = \cos \theta$

$\cos^2 \theta - \cos \theta = 0$

$\cos \theta (\cos \theta - 1) = 0$

$\cos \theta = 0$ or $\cos \theta = 1$

$\theta = 90^\circ, 270^\circ$ or $\theta = 0^\circ, 360^\circ$

$\theta = 0^\circ, 90^\circ, 270^\circ, 360^\circ$ (3)

(ii) $\sqrt{3} \sin \theta - 3 \cos \theta = 0$

$\sqrt{3} \sin \theta = 3 \cos \theta$

$\tan \theta = \sqrt{3}$

$Q1, 3$

$\tan \alpha = \sqrt{3}$

$\alpha = 60^\circ$

$\therefore \theta = \alpha, 180 + \alpha$

$\theta = 60^\circ, 240^\circ$ (3)

(iii) $\sin 3\theta = \frac{1}{2}$

$Q1, 2$

$\sin \alpha = \frac{1}{2}$

$3\theta = \alpha, 180 - \alpha$ $0^\circ \leq 3\theta \leq 1080^\circ$

$3\theta = 30^\circ, 150^\circ, 390^\circ, 510^\circ, 750^\circ, 870^\circ$

$\theta = 10^\circ, 50^\circ, 130^\circ, 170^\circ, 250^\circ, 290^\circ$ (3)

(iv) $4 \tan \theta + 1 = \tan^2 \theta = \sec^2 \theta$

$4 \tan \theta + 1 - \tan^2 \theta = 1 + \tan^2 \theta$

$2 \tan^2 \theta - 4 \tan \theta = 0$

$2 \tan \theta (\tan \theta - 2) = 0$

$\tan \theta = 0$ or $\tan \theta = 2$

$\theta = 0^\circ, 180^\circ, 360^\circ$

$Q1, 3$

$\tan \alpha = 2$

$\alpha = 63^\circ$

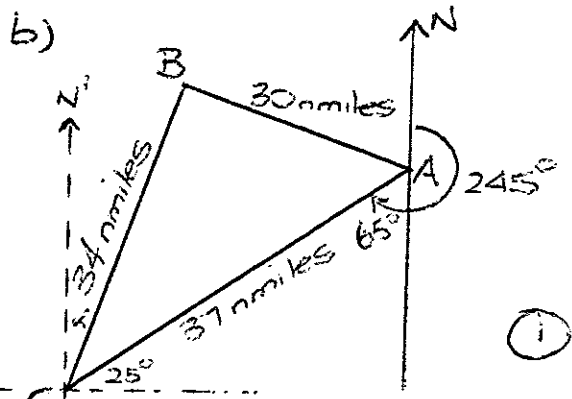
$\theta = \alpha, 180 + \alpha$

$\theta = 63^\circ, 243^\circ$

$\theta = 0^\circ, 63^\circ, 180^\circ, 243^\circ, 360^\circ$ (3)

Question 4

a) $(a^2 - b^2)^2 - (a - b)^4$
 $= (a + b)^2 (a - b)^2 - (a - b)^4$
 $= (a - b)^2 [(a + b)^2 - (a - b)^2]$
 $= (a - b)^2 (a + b + a - b)(a + b - a + b)$
 $= (a - b)^2 (2a)(2b)$
 $= 4ab(a - b)^2$ (3)



(ii) $\cos C = \frac{34^2 + 37^2 - 30^2}{2 \times 34 \times 37}$

$C = 50^\circ$ (2)

(iii) $\alpha + 30^\circ + 25^\circ = 90^\circ$

$\alpha = 35^\circ$

$\therefore$ bearing is 035° (2)

c) $\frac{1}{2} \cdot AB \cdot AC \sin 40^\circ = 40$

$\frac{1}{2} AB^2 = \frac{40}{\sin 40^\circ}$

$AB^2 = 80 \operatorname{cosec} 40^\circ$

$AB = \sqrt{80 \operatorname{cosec} 40^\circ}$ (2)

(ii)

$\frac{BC}{\sin 40^\circ} = \frac{AB}{\sin 70^\circ}$

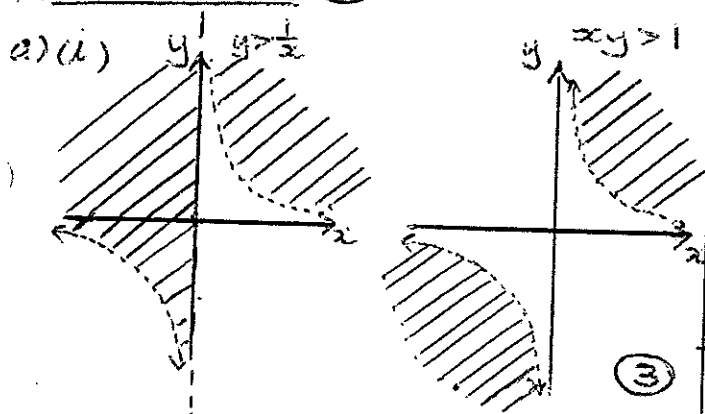
$BC = \frac{AB \sin 40^\circ}{\sin 70^\circ}$

$= \frac{\sqrt{80 \operatorname{cosec} 40^\circ} \cdot \sin 40^\circ}{\sin 70^\circ}$

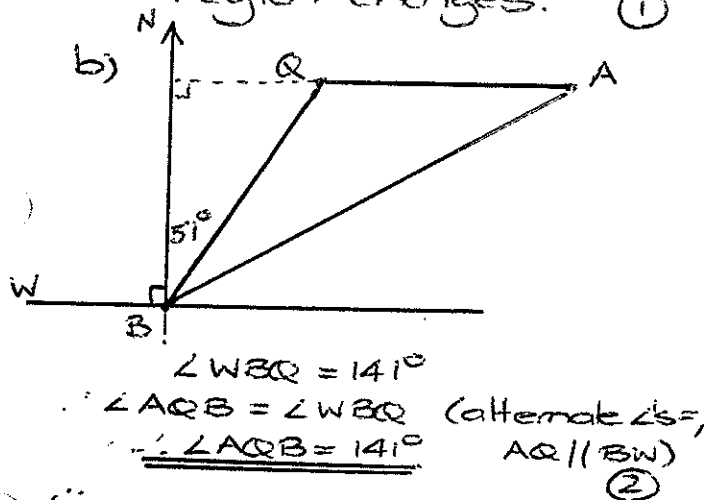
$= \frac{\sqrt{80 \sin 40^\circ}}{\sin 70^\circ}$

$= 7.6 \text{ cm}$ (3)

Question 5 (14)



(ii) The regions are not the same. They differ as to get $xy > 1$ you must multiply $y > \frac{1}{x}$ by x , but when $x < 0$ the inequality sign would change, thus the region changes. (1)



(ii) $\frac{AQ}{h} = \tan 78^\circ$
 $AQ = h \tan 78^\circ$ (2)

(iii) $BQ = h \tan 79^\circ$ (1)

(iv) $1000^2 = h^2 \tan^2 78^\circ + h^2 \tan^2 79^\circ - 2 \cdot h \tan 78^\circ \cdot h \tan 79^\circ \cos 141^\circ$

$1000^2 = h^2 (\tan^2 78^\circ + \tan^2 79^\circ - 2 \tan 78^\circ \tan 79^\circ \cos 141^\circ)$

$h^2 = \frac{1000^2}{\tan^2 78^\circ + \tan^2 79^\circ - 2 \tan 78^\circ \tan 79^\circ \cos 141^\circ}$

$h = \sqrt{\tan^2 78^\circ + \tan^2 79^\circ - 2 \tan 78^\circ \tan 79^\circ \cos 141^\circ}$

$h = 108 \text{ m}$ (3)

c) $\frac{1 + \cot \theta}{\operatorname{cosec} \theta} - \frac{1 + \tan \theta}{\sec \theta}$
 $= \frac{1 + \frac{\cos \theta}{\sin \theta}}{\frac{1}{\sin \theta}} - \frac{1 + \frac{\sin \theta}{\cos \theta}}{\frac{1}{\cos \theta}}$

$= \sin \theta + \cos \theta - (\cos \theta + \sin \theta)$
 $= 0$ (2)

Question 6

a) $\frac{1}{x} + \frac{3}{y} = 3$ (-)

$\frac{1}{x} + y = 1$

$\frac{3}{y} - y = 2$

$3 - y^2 = 2y$

$y^2 + 2y - 3 = 0$

$(y+3)(y-1) = 0$

$y = -3$ or $y = 1$

$y = 1, \frac{1}{x} + 1 = 1$

$\frac{1}{x} = 0$

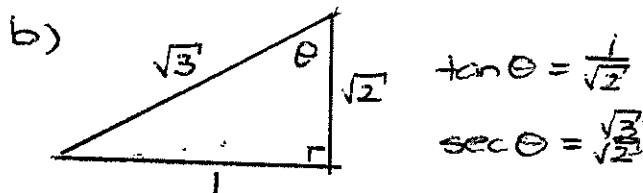
no solutions

$y = -3, \frac{1}{x} - 3 = 1$

$\frac{1}{x} = 4$

$x = \frac{1}{4}$

$\therefore x = \frac{1}{4}, y = -3$ (3)



$\frac{\tan \theta}{1 - \sec \theta} = \frac{\frac{1}{\sqrt{2}}}{1 - \frac{\sqrt{3}}{\sqrt{2}}}$

$= \frac{1}{\sqrt{2} - \sqrt{3}} \times \frac{\sqrt{2} + \sqrt{3}}{\sqrt{2} + \sqrt{3}}$

$= \frac{\sqrt{2} + \sqrt{3}}{\sqrt{2} - \sqrt{3}}$

$= -\sqrt{2} - \sqrt{3}$ (3)

(ii) If θ is obtuse $\tan \theta < 0$, $\sec \theta < 0$

$\frac{\tan \theta}{1 - \sec \theta} = \frac{-1}{\sqrt{2} + \sqrt{3}} \times \frac{\sqrt{2} - \sqrt{3}}{\sqrt{2} - \sqrt{3}}$

$= \sqrt{2} - \sqrt{3}$ (2)

$$\begin{aligned} \text{c) } \cos \theta &= \frac{\frac{q^2}{9} + r^2 - x^2}{2 \times \frac{q}{3} \times r} \\ &= \frac{q^2 + 9r^2 - 9x^2}{6qr} \quad (2) \end{aligned}$$

$$\begin{aligned} \text{(ii) In } \triangle QTR, \angle R &= 90^\circ - \theta \\ \cos(90^\circ - \theta) &= \frac{\frac{q^2}{9} + p^2 - y^2}{2 \times \frac{q}{3} \times p} \\ &= \frac{q^2 + 9p^2 - 9y^2}{6pq} \end{aligned}$$

$$\text{But } \cos(90^\circ - \theta) = \sin \theta$$

$$\therefore \sin \theta = \frac{q^2 + 9p^2 - 9y^2}{6pq}$$

$$\begin{aligned} \text{(iii) } \cos \theta &= \frac{r}{q} \\ \therefore \frac{r}{q} &= \frac{q^2 + 9r^2 - 9x^2}{6qr} \end{aligned}$$

$$6r^2 = q^2 + 9r^2 - 9x^2$$

$$9x^2 = q^2 + 3r^2$$

$$\sin \theta = \frac{p}{q}$$

$$\therefore \frac{p}{q} = \frac{q^2 + 9p^2 - 9y^2}{6pq}$$

$$6p^2 = q^2 + 9p^2 - 9y^2$$

$$9y^2 = q^2 + 3p^2$$

$$9(x^2 + y^2) = 2q^2 + 3p^2 + 3p^2$$

$$\text{But } p^2 + r^2 = q^2 \text{ (Pythagoras)}$$

$$\therefore 9(x^2 + y^2) = 2q^2 + 3q^2$$

$$= 5q^2 \quad (3)$$